

Network Scanning and Host Enumeration with Nmap

Internal Network Scan Report

- **Date:** 2026-02-21
- **Tool Used:** Nmap 7.98
- **Scope:** 192.168.0.0/24 subnet

```
jonathan@Jonathans-MacBook-Pro ~ % ipconfig getifaddr en0
192.168.0.15
jonathan@Jonathans-MacBook-Pro ~ % nmap -sn 192.168.0.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-21 01:30 -0400
Nmap scan report for 192.168.0.1
Host is up (0.010s latency).
Nmap scan report for 192.168.0.3
Host is up (0.0056s latency).
Nmap scan report for 192.168.0.11
Host is up (0.046s latency).
Nmap scan report for 192.168.0.14
Host is up (0.0087s latency).
Nmap scan report for 192.168.0.15
Host is up (0.00037s latency).
Nmap scan report for 192.168.0.252
Host is up (0.0043s latency).
Nmap done: 256 IP addresses (6 hosts up) scanned in 29.15 seconds
```

```
jonathan@Jonathans-MacBook-Pro ~ % nmap -sV -oN scan_results.txt 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-21 01:51 -0400
Nmap scan report for 192.168.0.1
Host is up (0.011s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE      SERVICE      VERSION
80/tcp    open      http         lighttpd
443/tcp   open      ssl/http     lighttpd
5000/tcp  open      upnp         MiniUPnP 1.9 (RHEL 6.10; UPnP 1.1)
8081/tcp  filtered  blackice-icecap
8082/tcp  filtered  blackice-alerts
Service Info: OS: Linux; CPE: cpe:/o:redhat:enterprise_linux:6.10
```

Findings

The network contains a small number of active devices relative to the available address space, suggesting no obvious signs of excessive or unexpected hosts. No hidden or non-responsive but suspicious devices were detected during the ping sweep.

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Out of 256 possible IP addresses in the /24 subnet, 6 hosts were identified as active:

- **192.168.0.1** – Default gateway (router)
- **192.168.0.3** – Active device
- **192.168.0.11** – Active device
- **192.168.0.14** – Active device
- **192.168.0.15** – Scanning host (local machine)
- **192.168.0.252** – Active device

A service version scan (-sV) was performed on **192.168.0.1**. The internal scan of 192.168.0.1 indicates a generally secure configuration with minimal exposed services.

The scan revealed:

- Only expected management ports (80 and 443) are open.
- Majority of ports (995) are closed.
- Firewall filtering is active on additional ports.

In conclusion, this types of scans help understand the network landscape and helps maintain a watchful eye on any unwanted devices or services.

Operating System Detection

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-21 03:30 -0400
Nmap scan report for 192.168.0.1
Host is up (0.0092s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
80/tcp    open      http
443/tcp   open      https
5000/tcp  open      upnp
8081/tcp  filtered  blackice-icecap
8082/tcp  filtered  blackice-alerts
MAC Address: 78:57:10:53:53:53 ( )
Device type: WAP|general purpose
Running: Ubee embedded, Arris embedded, Linux 2.6.X
OS CPE: cpe:/h:ubee:evw3226 cpe:/h:arris:tg1672 cpe:/h:arris:tg862g cpe:/o:linux:linux_kernel:2.6.18
OS details: Ubee EVW3226 or Arris TG1672 or TG862G cable modem (Linux 2.6.18)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.37 seconds
```

The **-O** scan can provide details like:

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- Target IP – 192.168.0.1
- Detected Operating System – Linux 2.6.X
- Device Type – General Purpose
- MAC address - xx:xx...
- Network Distance – 1 hop

TCP SYN SCAN

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-21 03:48 -0400
Nmap scan report for 192.168.0.1
Host is up (0.036s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
80/tcp    open       http
443/tcp   open       https
5000/tcp  open       upnp
8081/tcp  filtered   blackice-icecap
8082/tcp  filtered   blackice-alerts
MAC Address: [REDACTED]

Nmap done: 1 IP address (1 host up) scanned in 2.96 seconds
```

A TCP SYN scan (**-sS**) was performed to identify open ports on the target system. This technique sends SYN packets to initiate a TCP handshake but does not complete the connection, allowing for efficient and relatively stealthy port discovery. During the scan the ports open are the ones to be expected and there are no unusual open ports.

UDP Scan

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-21 04:03 -0400
Stats: 0:03:09 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 19.04% done; ETC: 04:20 (0:13:23 remaining)
Stats: 0:11:24 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 65.47% done; ETC: 04:21 (0:06:01 remaining)
Nmap scan report for 192.168.0.1
Host is up (0.0069s latency).
Not shown: 998 closed udp ports (port-unreach)
PORT      STATE      SERVICE
67/udp    open|filtered dhcps
1900/udp  open       upnp
MAC Address: [REDACTED]

Nmap done: 1 IP address (1 host up) scanned in 1088.15 seconds
```

A UDP scan (**-sU**) was conducted to identify exposed UDP services. Due to the connectionless nature of UDP, results were interpreted based on ICMP responses and service replies. UDP scanning is slower but essential for identifying services such as DNS, SNMP, and NTP.

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The scan revealed:

- 998 UDP ports closed (ICMP port unreachable responses received)
- Port 67/udp (DHCP) – open|filtered
- Port 1900/udp (UPnP) – open

The DHCP service is expected behavior, as the router functions as the network's DHCP server. Port 1900 indicates that UPnP is enabled for internal device discovery and automatic port forwarding. Overall, the UDP exposure appears minimal and consistent with standard router functionality.

Lightweight Vulnerability Scan

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```
Stats: 0:02:08 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.67% done; ETC: 04:35 (0:00:00 remaining)
Nmap scan report for 192.168.0.1
Host is up (0.044s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
80/tcp    open       http
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-sql-injection:
|_  Possible sqlmap for queries:
|_  http://192.168.0.1:80/crypt.js?ver=9.1.103DV8W%27%20OR%20sqlspider
|_  http://192.168.0.1:80/text.js?ver=9.1.103DV8W%27%20OR%20sqlspider
443/tcp    open       https
|_http-sql-injection:
|_  Possible sqlmap for queries:
|_  https://192.168.0.1:443/crypt.js?ver=9.1.103DV8W%27%20OR%20sqlspider
|_  https://192.168.0.1:443/text.js?ver=9.1.103DV8W%27%20OR%20sqlspider
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
ssl-dh-params:
VULNERABLE:
Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
State: VULNERABLE
Transport Layer Security (TLS) services that use anonymous
Diffie-Hellman key exchange only provide protection against passive
eavesdropping, and are vulnerable to active man-in-the-middle attacks
which could completely compromise the confidentiality and integrity
of any data exchanged over the resulting session.
Check results:
ANONYMOUS DH GROUP 1
Cipher Suite: TLS_DH_anon_WITH_AES_256_CBC_SHA
Modulus Type: Non-safe prime
Modulus Source: RFC5114/1024-bit DSA group with 160-bit prime order subgroup
Modulus Length: 1024
Generator Length: 1024
Public Key Length: 1024
References:
https://www.ietf.org/rfc/rfc2246.txt

Diffie-Hellman Key Exchange Insufficient Group Strength
State: VULNERABLE
Transport Layer Security (TLS) services that use Diffie-Hellman groups
of insufficient strength, especially those using one of a few commonly
shared groups, may be susceptible to passive eavesdropping attacks.
Check results:
WEAK DH GROUP 1
Cipher Suite: TLS_DHE_RSA_WITH_AES_256_CBC_SHA
Modulus Type: Non-safe prime
Modulus Source: RFC5114/1024-bit DSA group with 160-bit prime order subgroup
Modulus Length: 1024
Generator Length: 1024
Public Key Length: 1024
References:
https://weakdh.org
|_http-aspnet-debug: ERROR: Script execution failed (use -d to debug)
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
5000/tcp   open       upnp
8081/tcp   filtered  blackice-icecap
8082/tcp   filtered  blackice-alerts
```

A lightweight vulnerability scan was performed against the default gateway (192.168.0.1).

Findings

1. Anonymous Diffie-Hellman Cipher Support

- TLS service supports anonymous DH cipher suites.
- This configuration does not provide authentication and may allow man-in-the-middle attacks.
- Risk: Medium (internal network context)

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2. Weak Diffie-Hellman Group (1024-bit)

- TLS configuration supports weak 1024-bit DH parameters.
- This is considered cryptographically insufficient by modern standards.
- Risk: Low–Medium (depending on exposure)

3. HTTP SQL Injection Checks

- Automated script indicated possible SQL injection in static JavaScript resources.
- Likely false positive due to reflection behavior.
- No confirmed exploitable injection identified.

Reccomendations

- Check router firmware update
- Disable remote administration
- Ensure router admin not exposed to WAN
- If firmware allows: disable weak ciphers